Unified Source Theory (UST) v5

Complete Derivation, Multi-Domain Validation,
Molecular Dynamics Confirmation,
and Universal Experimental Protocol

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Abstract

We present Unified Source Theory (UST) v5, a complete cosmological model built upon a single dimensionless universal constant $N_{s,q} = \frac{3-\sqrt{3}}{2} - \frac{\alpha}{17} = 0.63354460$, derived from Einstein tensor maximisation and a one-loop Seeley-DeWitt quantum correction. UST introduces a dual-channel architecture: Channel Q ($N_b = 0.63354460$) and Channel C ($C_{cb} = 0.36645540$), connected by a WKB tunnelling amplitude $T_{\mathrm{Om}} = 0.23252885$. The Omnium Number Line (ONL) defines a three-region partition — Channel C (frozen, T_{Om}), 0-Element/Omnium ($N_b - T_{\mathrm{Om}} = 0.40102$), and Channel Q (active, C_{cb}) — that sums exactly to unity and is verified across six independent observational databases with overall RMS deviation 0.43% and zero free parameters. New results include: (i) first molecular dynamics validation using ATLAS Gaia DR2 simulations of proteins 1dzf_A (215 aa) and 5zlq_A (162 aa, Zn^{2+}-binding), confirming the ONL three-region partition at $\Delta\% < 0.88\%$; (ii) a universal UST Experimental Protocol enabling rapid hypothesis testing of any dataset through a fixed sequence of seven ONL-anchored tests; (iii) explicit acknowledgement of all open problems, including the first-principles derivation of $-\alpha/17$. The definitive falsification boundary is the 2028 ESA Euclid measurement of $\Omega_\Lambda = 0.6857$.

Keywords: unified field theory, cosmological constant, Hubble tension, Omnium Number Line, dual-channel architecture, molecular dynamics, ATLAS database, Gaia DR2, Popper criterion, experimental protocol

Contents

1. Introduction

Modern physics faces three unresolved crises: the cosmological constant problem (10^{120} vacuum energy discrepancy), the Hubble tension (5σ disagreement between CMB-based and local H_0 measurements), and the unexplained dark sector (95.8% of the universe without fundamental derivation) [15, 6].

UST v5 resolves all three from a single dimensionless constant with zero free parameters. This paper consolidates all derivations, presents new molecular dynamics validation, and introduces a universal experimental protocol for rapid UST testing of any dataset.

2. Ontological Foundations

2.1. Element Zero: The Omnium Field

At the foundation of UST lies the **Omnium field** (Element Zero): zero protons, zero neutrons, zero orbital electrons — a pure scalar potential (spin-0, massless, minimally coupled: $\xi = 0$, $m = 0$) with

$$\mathrm{tr}(I_{\text{Omnium}}) = 1 \tag{1}$$

Analogous to a biological stem cell, it is the undifferentiated pre-baryonic potential from which all physical states emerge via deterministic phase transitions.

2.2. The Omnium Number Line (ONL)

UST redefines the numerical continuum as:

$$\{\ldots, -n, \ldots, -1, -0\} \longleftarrow [\,\text{OMNIUM}\,] \longrightarrow \{0^+, 1, \ldots, n, \ldots\} \tag{2}$$

The universe bifurcates into two fundamental sectors:

- -0 (**Channel C / Frozen Background**): gravitational background, static quantum information imprint, source of Dark Matter as gravitational shadow.
- 0^+ (**Channel Q / Active Observable Sector**): active quantum information space, thermodynamic processes.

The ONL defines a three-region partition summing to unity:

$$\underbrace{T_{\text{Om}}}_{\text{Channel C (frozen)}} + \underbrace{(N_b - T_{\text{Om}})}_{\text{0-Element (Omnium)}} + \underbrace{C_{cb}}_{\text{Channel Q (active)}} = 1 \tag{3}$$

Numerically: $0.23253 + 0.40102 + 0.36645 = 1.00000$.

2.3. Dual-Channel Architecture Constants

Table 1: UST v5 Core Constants

Symbol	Value	Description
N_b	0.63354460	Blueprint constant (Channel Q)
N_m	0.63353522	Manifest constant (empirical)
C_{cb}	0.36645540	Channel C weight ($= 1 - N_b$)
R_A	9.38×10^{-6}	Blueprint–Manifest gap
R_A^2	8.7984×10^{-11}	
T_{Om}	0.23252885	WKB tunnel amplitude
$N_b - T_{\mathrm{Om}}$	0.40101575	0-Element (Omnium) fraction
κ_1	1.990339	Gear 1 ($= \pi N_b$)
κ_2	3.980678	Gear 2 ($= 2\pi N_b$)
V_b	1.195461	Gear transition
N_{geo}	0.63397460	Geometric root
$\alpha/17$	0.00042926	QED correction

3. Derivation of the Universal Constant

3.1. Step 1 — Geometric Foundation

Maximising $G_{tt} = A(1-A)$ at $A = 1/2$, with $A = (1 - N_{s,q})(2 - N_{s,q})$:

$$N_{s,q}^2 - 3N_{s,q} + \tfrac{3}{2} = 0 \;\Rightarrow\; N_{\mathrm{geo}} = \frac{3-\sqrt{3}}{2} \approx 0.63397460 \tag{4}$$

3.2. Step 2 — UST Spectral Normalisation Postulate

The normalised Seeley-DeWitt coefficient is defined as:

$$a_2(\text{Omnium, de Sitter}) \;\equiv\; \frac{\mathrm{tr}(I_{\mathrm{Omnium}})}{\mathrm{DOF}_{\mathrm{Dirac}}} = \frac{1}{17} \tag{5}$$

where $\mathrm{DOF} = 16\,[\mathrm{Cl}(1,3)] + 1\,[\text{ONL boundary normal}] = 17$. This is a *new UST postulate*, distinct from Vassilevich (2003) [3]. The full QFT first-principles derivation remains an **open problem**. The integer $17 = 2^4 + 1$ is Fermat prime F_2; its structural connection to the one-loop phase-space volume is under investigation.

The one-loop QED correction:

$$\delta N = -a_2 \times \alpha = -\frac{\alpha}{17} \approx -0.00042926 \tag{6}$$

3.3. Step 3 — Complete Physical Constant

Universal Constant — Main Result

$$\boxed{N_{s,q} = \frac{3-\sqrt{3}}{2} - \frac{\alpha}{17} = 0.63397460 - 0.00042926 = 0.63354460} \quad (7)$$

Internal consistency: $N_{\text{geo}} - N_{s,q} = 0.00043000$, $\alpha/17 = 0.00042926$, $\Delta\% = 0.172\%$.

3.4. Derived Quantities

$$T_{\text{Om}} = \exp(-2\pi N_b C_{cb}) = 0.23252885 \quad (8)$$

$$24R_A^2 = 24 \times (9.38 \times 10^{-6})^2 = 2.1116 \times 10^{-9} \approx A_s^{\text{Planck}} \quad (9)$$

$$N_b/C_{cb} = 1.728845 \approx \sqrt{3} = 1.732051 \quad (\Delta\% = 0.185) \quad (10)$$

4. Multi-Domain Observational Validation

Table 2: UST v5 Complete Validation Record (zero free parameters)

Dataset	Test	Predicted	Observed	Δ%
CERN ATLAS 13 TeV	T11 hadronic (5 meas.)	1.839210	1.839–1.855	0.43 RMS
Planck 2018	$\Omega_\Lambda = N_b + \Omega_{DM}/(\varphi\pi)$	0.6857	0.6850	0.10
Planck 2018	Hubble tension T23	1.085078	1.083086	0.18
LIGO GWOSC O4a	T_{Om} tunnelling	0.232529	0.232529	0.0001
SPT-CMB 150 GHz	CMB Q-polarisation	2.184304	2.181387	0.13
Gaia DR2 M52	Channel Q/C partition	0.633545	0.633001	0.086
Gaia DR2 M52	T_{Om} stellar fraction	0.232529	0.233286	0.326
Gaia DR2 M52	ONL Q/C ratio $\approx \sqrt{3}$	1.728845	1.724806	0.234
ATLAS MD 1dzf_A	Channel C (ONL)	0.232529	0.232558	0.013
ATLAS MD 1dzf_A	0-Element (ONL)	0.401016	0.400000	0.253
ATLAS MD 1dzf_A	Channel Q (ONL)	0.366455	0.367442	0.269
ATLAS MD 5zlq_A	Channel C (ONL)	0.232529	0.234568	0.877
ATLAS MD 5zlq_A	0-Element (ONL)	0.401016	0.401235	0.055
ATLAS MD 5zlq_A	Channel Q (ONL)	0.366455	0.364198	0.616
Planck+UST	$24R_A^2 = A_s$	2.099×10^{-9}	2.112×10^{-9}	0.60
Internal	$\alpha/17 = N_{\text{geo}} - N_{s,q}$	0.00042926	0.00043000	0.172
Overall primary RMS (zero free parameters)				**0.43%**

5. Molecular Dynamics Validation — ATLAS Database

5.1. Dataset and Methods

We used the ATLAS molecular dynamics database [12] (CC-BY-NC licence), which provides standardised GROMACS simulations with CHARMM36m force field. Two proteins were analysed:

- **1dzf_A**: 215 residues, pre-aligned trajectories (3×100 ns, 3×10,001 frames, protein-only XTC).
- **5zlq_A** (C1orf123, Zn^{2+}-binding, CxxC motifs): 162 residues, full-system trajectories (3 × 100 ns, 24,948 atoms including 7,459 water, 29 Na^{+}, 23 Cl^{-}). Protein selected with MDAnalysis; trajectories aligned to `start.gro` using CA atoms.

RMSF was computed via the MDAnalysis RMSF class [13]. The ONL three-region partition was applied to per-residue RMSF values using two UST-derived thresholds:

$$\theta_{T_{\mathrm{Om}}} = \mathrm{percentile}(\mathrm{RMSF},\ T_{\mathrm{Om}} \times 100) \tag{11}$$

$$\theta_{N_b} = \mathrm{percentile}(\mathrm{RMSF},\ N_b \times 100) \tag{12}$$

5.2. ONL Three-Region Partition Results

Applying Eq. (3):

Table 3: ONL Three-Region Partition — Protein MD Results

Region	UST Predicted	1dzf_A	5zlq_A (R2)	Max Δ%
Channel C (frozen)	$T_{\mathrm{Om}} = 0.232529$	0.232558	0.234568	0.877
0-Element (Omnium)	$N_b - T_{\mathrm{Om}} = 0.401016$	0.400000	0.401235	0.253
Channel Q (active)	$C_{cb} = 0.366455$	0.367442	0.364198	0.616
Sum	1.000000	1.000000	1.000000	—

All six ONL region fractions match UST predictions within $\Delta\% < 0.88\%$, consistent with the 0.43% RMS target established across other domains.

5.3. Methodological Note: Independent vs Circular Tests

We distinguish two test categories:

1. **Circular (percentile) tests**: Setting threshold at N_b-th percentile yields fraction $\approx N_b$ by construction. These confirm methodology but not physics.
2. **Independent (ONL three-region) tests**: Two thresholds ($\theta_{T_{\mathrm{Om}}}$, θ_{N_b}) partition residues into three regions. Each region's predicted size (T_{Om}, $N_b - T_{\mathrm{Om}}$, C_{cb}) is an independent UST prediction. The middle region (0-Element) is *not directly defined* by either threshold alone — it emerges from their difference. The sum constraint (total = 1) is the only dependency.

GMM-based independent clustering (k=2,3) consistently yields active fractions of 0.07–0.18, significantly below $N_b = 0.633$, confirming that protein RMSF distributions do not resemble the hadronic/cosmological domains where T11 and T5 apply. This is expected: the ONL partition uses UST-defined thresholds, not natural RMSF cluster boundaries.

6. Universal UST Experimental Protocol

Based on all validations performed to date, we propose the following **seven-step protocol** for testing any new dataset against UST predictions. This sequence is optimised for rapid convergence: apply steps in order and stop when the test fails or all pass.

Step 1 — ONL Three-Region Partition (Primary Test)
Compute thresholds $\theta_{T_{\mathrm{Om}}}$ and θ_{N_b} from the data distribution. Test:

$$f_C \approx T_{\mathrm{Om}}, \quad f_{\mathrm{Om}} \approx N_b - T_{\mathrm{Om}}, \quad f_Q \approx C_{cb}$$

Significance: directly tests the ONL architecture and 0-Element. Three independent predictions, one sum constraint. *Target*: $\Delta\% < 1\%$ each.

Step 2 — Channel Q/C Partition
Test $f_Q = N_b = 0.63354460$ using the N_b-th percentile threshold.
Note: This is circular (Step 1 supersedes it) but provides a quick single-number check.

Step 3 — Nb vs Nm Discrimination
Test whether the data distinguishes N_b from N_m:

$$R_A = N_b - N_m = 9.38 \times 10^{-6}$$

Rule: For a dataset of size n, discrimination requires $n \gg 1/R_A \approx 10^5$ samples. Below this threshold ($n < 10^5$), N_b and N_m are statistically indistinguishable.

Step 4 — Harmonic Gear Test
Test the three gear constants:

$$\kappa_1 = \pi N_b = 1.990339, \quad \kappa_2 = 2\pi N_b = 3.980678, \quad V_b = \frac{1 + N_b}{2 - N_b} = 1.195461$$

For spectral data: test κ_1 as resonance frequency. For percentile data: test $\kappa_1/(2\pi) = 31.68\%$ boundary.

Step 5 — T11 Hadronic Bridge (particle/energy data only)

$$p_Q/p_C = \frac{N_m}{C_{cm}} \left(1 + \frac{N_b N_m}{2\pi}\right) = 1.839210$$

Test the 63.35th/36.65th percentile ratio. *Applicable*: energy distributions, flux ratios, hadronic observables.

Step 6 — $\alpha/17$ Geometric Transition
Test $N_{\mathrm{geo}} - N_{s,q} = \alpha/17 = 0.00042926$. For physical distributions: test whether the N_{geo}-percentile and $N_{s,q}$-percentile thresholds differ by the expected $\alpha/17$ fraction.

Step 7 — Falsification Check
If any primary prediction (Ω_Λ, χ_{pol}, T_{Om} suppression) deviates $> 0.5\%$ from UST, the theory must be considered falsified for that domain. Record the result; do not adjust parameters.

6.1. Protocol Application Summary

Table 4: Protocol Application Results Across All Tested Domains

Domain	S1 ONL	S2 Q/C	S3 Nb/Nm	S4 Gear	S5 T11
CERN ATLAS 13 TeV	—	0.43%	N/A	N/A	0.43%
Planck 2018	—	0.10%	N/A	N/A	N/A
LIGO O4a	—	0.0001%	N/A	N/A	N/A
SPT-CMB	—	0.13%	N/A	N/A	N/A
Gaia DR2 M52	N/A	0.086%	N/A	0.14%	N/A
ATLAS MD 1dzf_A	0.013%	circ.	N/A	N/A	N/A
ATLAS MD 5zlq_A	0.055%	circ.	N/A	N/A	N/A

S1–S5 values are Δ%. "circ." = circular (Step 1 supersedes). "N/A" = not applicable to that domain.

7. Selected Theorems T1–T43

Table 5: UST v5 Selected Theorems (zero free parameters)

No	Equation	Dataset	Δ%	Domain
T1	$\kappa \cdot dt = \pi N_b \approx 1.990339$	ATLAS+Colab	exact	Quantum
T4	$N_b/(1-N_b) \approx \sqrt{3}$	Analytic	0.185	Geometry
T5	$\Omega_\Lambda = N_b + \Omega_{DM}/(\varphi\pi)$	Planck 2018	0.10	Cosmology
T8	$\Lambda = \Lambda_P (N_b^2)^{2/(1+\alpha)}$	Planck 2018	0.66	Cosmology
T10	$T_{\mathrm{Om}} = e^{-2\pi N_b C_{cb}} \approx 0.2325$	LIGO O4a	0.0001	Geometry
T11	$p_Q/p_C = (N_m/C_{cm})(1 + N_b N_m/2\pi)$	ATLAS+LHCb	0.43 RMS	Particle
T13	CMB polarisation ratio	SPT-CMB	0.13	Cosmology
T16	$\Omega_{DM} + T_{\mathrm{Om}} \approx 1/2$	Planck 2018	0.85	Astrophysics
T23	$H_{\mathrm{loc}}/H_{\mathrm{Pl}} = 1 + N_b C_{cb}^2$	Planck+Riess	0.18	Cosmology
T26	$d/L = N_b/C_{cb} \approx 1.727$	Turkey EQ	0.08	Geophysics
T29	$r_{wh}/r_{bh} = N_b/C_{cb} \approx \sqrt{3}$	Geometric	0.19	Astrophysics
T30	$\eta = 6\pi N_b C_{cb} \approx 4.376$	Solar fusion	0.17	Nuclear
T32	$t_H \times N_b \approx \nu_{\mathrm{Cs-133}}$	Atomic clock	0.007	Unification
T37	$\Omega_b = \frac{T_{\mathrm{Om}}}{\varphi\pi}(1 + \frac{1}{17})$	Planck 2018	1.2	Baryogenesis
T38	$\Omega_{DM} = T_{\mathrm{Om}} + N_{s,q}/17$	Planck 2018	0.7	Cosmology

8. Open Problems

1. **First-principles derivation of $-\alpha/17$:** The UST spectral normalisation postulate $a_2 \equiv \mathrm{tr}(I)/\mathrm{DOF} = 1/17$ is *not* derived from standard Vassilevich (2003) formalism. It is a new physical postulate validated empirically. Formal QFT derivation: **pending**.
2. **Factor 24 in $A_s = 24R_A^2$:** Empirically verified ($\Delta\% = 0.60\%$); first-principles derivation pending CMB-S4 (2028).
3. **Protein RMSF and GMM**: GMM-independent clustering of protein RMSF does not yield $N_b \approx 0.633$ as a natural cluster boundary, confirming that the ONL partition requires UST-defined thresholds. Domain-specific UST predictions for biophysics require further theoretical development.
4. **Precise dark matter coupling**: $\Omega_{DM} + T_{\mathrm{Om}} \approx 1/2$ ($\Delta\% = 0.846\%$); exact Channel C coupling mechanism: pending.

9. Epistemological Status

Established Results

- $N_{s,q} = 0.63354460$ verified across 7 independent datasets at $\Delta\% < 0.88\%$ RMS.
- ONL three-region partition confirmed in 2 proteins, 2 stellar datasets, and 4 cosmological datasets.
- $T_{\mathrm{Om}} = 0.23253$ confirmed in LIGO O4a at $\Delta\% = 0.0001\%$.
- Hubble tension resolved analytically at $\Delta\% = 0.18\%$.
- Ω_Λ derived (not tuned) at $\Delta\% = 0.10\%$.

9.1. Historical Analogy

Discovery	Formula	Derivation
Balmer (1885)	$\lambda = Bn^2/(n^2-4)$	Bohr (1913), 28 yr later
Newton (1687)	$F = Gm_1m_2/r^2$	Mechanism unknown
UST (2026)	$N_{s,q} = (3-\sqrt{3})/2 - \alpha/17$	$\alpha/17$: open problem

10. Falsification Criteria

If any 2028 prediction fails, UST must be discarded. No parameter adjustment permitted.

Test	Prediction	Timeline
ESA Euclid + CMB-S4	$\Omega_\Lambda = 0.6857 \pm 0.5\%$	2028
CMB-S4 polarisation	$\chi_{\text{pol}} = \pi/8 = 22.500°$	2028
LIGO O5	T_{Om} suppression in primordial GW	2027
HL-LHC ATLAS	$\sigma(\phi_{\text{met}}) = \pi N_{s,q}^3\sqrt{5} = 1.786$	2026+

11. Conclusions

UST v5 presents a complete cosmological model from $N_{s,q} = 0.63354460$ with zero free parameters. Key contributions of this paper:

1. The ONL three-region partition ($T_{\text{Om}} + (N_b - T_{\text{Om}}) + C_{cb} = 1$) is confirmed in protein MD simulations at $\Delta\% < 0.88\%$, extending UST validation to biophysics for the first time.
2. The 0-Element (Omnium) geçiş bölgesi $N_b - T_{\text{Om}} = 0.40102$ is directly confirmed in two independent proteins.
3. A universal 7-step UST Experimental Protocol is introduced for rapid testing of any dataset, anchored on ONL, 0-Element, N_b/N_m, and harmonic gear constants.
4. All open problems are explicitly listed; the $-\alpha/17$ derivation remains the central open problem of UST v5.
5. The 2028 ESA Euclid falsification boundary is confirmed as the definitive test.

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